

I.2 — Albanese Varieties and Albanese Mappings

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Yi Li

Big picture

The aim of this note is to give a self-contained introduction to Albanese varieties and Albanese mappings, with a view towards minimal model theory and the classification of certain varieties. We first give an elementary introduction to Albanese varieties and the Albanese mapping. We then turn to two research topics: the existence of good minimal models for varieties with maximal Albanese dimension ([Section 5](#)), and Demailly–Peternell–Schneider’s classification of compact Kähler manifolds with nef tangent bundle ([Section 4](#)). We end the note by pointing out some further directions.

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1 Albanese varieties

1.1 Construction over a compact Kähler manifold

Let us first construct the Albanese torus. The following observation is obvious: on a compact Kähler manifold holomorphic forms are automatically closed.

Lemma 1.1. *Let X be a compact Kähler manifold with Kähler metric ω . If $\alpha \in H^0(X, \Omega_X^p)$ is a holomorphic p -form, then α is closed.*

Proof. By definition $\bar{\partial}\alpha = 0$. By type reasons the formal adjoint satisfies $\bar{\partial}^*\alpha = 0$ as well, so $\Delta_{\bar{\partial}}\alpha = 0$. The Kähler identities give $\Delta_{\bar{\partial}} = \Delta_{\partial}$, so $\Delta_{\partial}\alpha = 0$. Then

$$0 = (\Delta_{\partial}\alpha, \alpha) = \|\partial\alpha\|^2 + \|\bar{\partial}^*\alpha\|^2 \implies \partial\alpha = 0.$$

Together with $\bar{\partial}\alpha = 0$ this gives $d\alpha = 0$. ■

The following proposition is important for the construction of Albanese torus for compact Kähler manifold.

Proposition 1.2. *Let X be a compact Kähler manifold with Kähler form ω .*

(1) *There is a well-defined homomorphism*

$$\varphi : H_1(X, \mathbb{Z}) \rightarrow (H^0(X, \Omega_X^1))^{\vee}, \quad [\gamma] \mapsto \left(\alpha \mapsto \int_{\gamma} \alpha \right).$$

(2) *The image of φ is a lattice in $(H^0(X, \Omega_X^1))^{\vee}$; hence the quotient is a complex torus of dimension $\dim_{\mathbb{C}} H^0(X, \Omega_X^1)$.*

Proof of (1). By Lemma 1.1 every $\alpha \in H^0(X, \Omega_X^1)$ is closed. If $[\gamma] = [\gamma']$ in $H_1(X, \mathbb{Z})$, there is a singular 2-chain S with $\gamma - \gamma' = \partial S$, so by Stokes' theorem

$$\int_{\gamma - \gamma'} \alpha = \int_{\partial S} \alpha = \int_S d\alpha = 0.$$

Hence $\varphi([\gamma])$ depends only on the homology class $[\gamma]$. ■

Proof of (2). Our proof follows [Huy05, page 135]. We must show that $H_1(X, \mathbb{Z})$ maps under φ to a lattice in $(H^0(X, \Omega_X^1))^{\vee}$. The natural map

$$H^{2n-1}(X, \mathbb{Z}) \rightarrow H^{2n-1}(X, \mathbb{C})$$

has image a lattice, since by the universal coefficient theorem tensoring with $\mathbb{Q}$ kills torsion. Applying Serre duality (SD) and Poincaré duality (PD) gives a commutative diagram

$$\begin{array}{ccc} H^0(X, \Omega_X^1)^{\vee} & \xleftarrow[\cong]{\text{SD}} & H^n(X, \Omega_X^{n-1}) \\ \uparrow \varphi & & \uparrow p \\ H_1(X, \mathbb{Z}) & \xrightarrow[\cong]{\text{PD}} & H^{2n-1}(X, \mathbb{Z}) \end{array}$$

where p is the composite $H^{2n-1}(X, \mathbb{Z}) \rightarrow H^{2n-1}(X, \mathbb{C}) \rightarrow H^{n-1, n}(X)$, whose image is again a lattice (tensoring with $\mathbb{C}$ kills torsion).

The image of φ agrees with the image of p : by the definition of Poincaré duality,

$\int_X \omega \wedge \text{PD}([\gamma]) = \int_\gamma \omega$, while by the definition of Serre duality

$$\text{SD}(\text{PD}([\gamma]))(\omega) := \int_X \text{PD}([\gamma]) \wedge \omega = \int_\gamma \omega = \varphi([\gamma])(\omega).$$

Hence $\text{im}(\varphi) = \text{im}(p)$ is a lattice, of full rank $2 \dim_{\mathbb{C}} H^0(X, \Omega_X^1)$. ■

Remark 1.3 (Lattices). A lattice in a complex vector space V of dimension g is a discrete subgroup $\Lambda \subset V$ of maximal rank, i.e. a free abelian group of rank $2g$; the quotient V/Λ is then a complex torus of dimension g .

Definition 1.4 (Albanese torus, first version). Let X be a compact Kähler manifold. The complex torus

$$\text{Alb}(X) := (H^0(X, \Omega_X^1))^\vee / H_1(X, \mathbb{Z})$$

is called the Albanese variety (or Albanese torus) of X ; by Proposition 1.2 it is well defined. Fixing a base point $x_0 \in X$ produces the Albanese mapping (Definition 2.1 below).

2 The Albanese mapping

2.1 Construction of Albanese mapping

Definition 2.1 (Albanese mapping, [Huy05, Definition 3.3.7]). Let X be a compact Kähler manifold, fix $x_0 \in X$, and let $\omega_1, \dots, \omega_q$ be a basis of $H^0(X, \Omega_X^1)$. The Albanese mapping of X is

$$\text{alb}_X : X \rightarrow \text{Alb}(X), \quad x \mapsto \left(\int_{x_0}^x \omega_1, \dots, \int_{x_0}^x \omega_q \right) \bmod H_1(X, \mathbb{Z}).$$

Remark 2.2. Since X is compact, the Albanese mapping is proper for trivial reasons. Unfortunately, the Albanese mapping need not be projective in general.

Proposition 2.3. *The Albanese mapping is well defined, i.e., it does not depend on the choice of path from x_0 to x , and a different choice of base point translates alb_X by a constant.*

Proof. Fix the basis $\omega_1, \dots, \omega_q$ of $H^0(X, \Omega_X^1)$ used to define alb_X . By Lemma 1.1, each ω_i is closed.

We first show that it is independent of the path. Let $x \in X$ and let γ, γ' be two paths from

x_0 to x . Then $\sigma := \gamma \cdot (\gamma')^{-1}$ is a closed loop at x_0 , and for each i ,

$$\int_{\gamma} \omega_i - \int_{\gamma'} \omega_i = \int_{\gamma \cdot (\gamma')^{-1}} \omega_i = \int_{\sigma} \omega_i.$$

Hence

$$\left(\int_{\gamma} \omega_1, \dots, \int_{\gamma} \omega_q \right) - \left(\int_{\gamma'} \omega_1, \dots, \int_{\gamma'} \omega_q \right) = \varphi([\sigma]) \in \Lambda.$$

Since $\text{alb}_X(x)$ is by definition the class of $(\int_{x_0}^x \omega_1, \dots, \int_{x_0}^x \omega_q)$ in $\mathbb{C}^q/\Lambda$, the two choices of path yield vectors in $\mathbb{C}^q$ differing by an element of Λ , hence define the same class in $\mathbb{C}^q/\Lambda$. Thus $\text{alb}_X(x)$ does not depend on the path chosen, and alb_X is a well-defined map $X \rightarrow \text{Alb}(X)$. ■

2.2 Universal property of Albanese mapping

The Albanese mapping enjoys a universal property.

Theorem 2.4 (Universal property of the Albanese map, [Huy05]). *Let $\varphi : X \rightarrow T$ be a holomorphic map from a compact complex manifold X (with base point x_0) to a complex torus T . There is a unique homomorphism of complex tori $\tilde{\varphi} : \text{Alb}(X) \rightarrow T$ such that*

$$\varphi(x) = \tilde{\varphi}(\text{alb}_X(x)) + \varphi(x_0), \quad x \in X.$$

$$\begin{array}{ccc} X & \xrightarrow{\text{alb}_X} & \text{Alb}(X) \\ & \searrow \varphi & \downarrow \tilde{\varphi} + \varphi(x_0) \\ & & T \end{array}$$

Proof. Write $T = \mathbb{C}^g/\Lambda$ for a lattice $\Lambda \subset \mathbb{C}^g$. Let $\pi : \tilde{X} \rightarrow X$ be the universal cover, with $\pi_1(X, x_0)$ acting as the deck transformation group, and fix a lift $\tilde{x}_0 \in \tilde{X}$ of x_0 . Recall

$$\text{Alb}(X) = H^0(X, \Omega_X^1)^\vee / H_1(X, \mathbb{Z}), \quad \text{alb}_X(x) = \left(\omega \mapsto \int_{x_0}^x \omega \right) \bmod H_1(X, \mathbb{Z}).$$

Since $\tilde{X}$ is simply connected, the map $\varphi \circ \pi : \tilde{X} \rightarrow T$ lifts along the universal cover $\mathbb{C}^g \rightarrow T$ to a holomorphic map

$$\phi = (\phi_1, \dots, \phi_g) : \tilde{X} \rightarrow \mathbb{C}^g, \quad \phi(\tilde{x}_0) = 0.$$

The lifting is equivariant. For $\gamma \in \pi_1(X, x_0)$, both $\phi \circ \gamma$ and ϕ lift $\varphi \circ \pi$ (as $\pi \circ \gamma = \pi$), so they differ by a deck transformation of $\mathbb{C}^g \rightarrow T$:

$$\phi(\gamma \cdot z) - \phi(z) \in \Lambda \quad \text{for all } z \in \tilde{X}. \quad (*)$$

Fix γ and set $f_\gamma := \phi \circ \gamma - \phi$, a holomorphic map $\tilde{X} \rightarrow \mathbb{C}^g$ valued in the discrete set Λ by (*). Continuity of f_γ then forces f_γ to be locally constant. As $\tilde{X}$ is connected, f_γ is a single constant $c_\gamma \in \Lambda$. Differentiating,

$$\gamma^*(d\phi_i) = d\phi_i, \quad i = 1, \dots, g, \quad \gamma \in \pi_1(X, x_0),$$

That is, $d\phi_i$ is a π_1 -invariant holomorphic one-form, so each $d\phi_i$ is the pullback $\pi^*\eta_i$ of a unique $\eta_i \in H^0(X, \Omega_X^1)$.

Fix a basis $\omega_1, \dots, \omega_q$ of $H^0(X, \Omega_X^1)$, $q = h^0(X, \Omega_X^1)$, and write $\eta_i = \sum_{j=1}^q a_{ij} \omega_j$. The matrix $A = (a_{ij})$ defines a linear map

$$A : H^0(X, \Omega_X^1)^\vee \rightarrow \mathbb{C}^g, \quad A(\xi)_i = \xi(\eta_i).$$

Every class in $H_1(X, \mathbb{Z})$ is represented by a loop γ at x_0 . Identifying $[\gamma]$ with the functional $\omega \mapsto \int_\gamma \omega$, and lifting γ to a path from $\tilde{x}_0$ to $\gamma \cdot \tilde{x}_0$ in $\tilde{X}$,

$$A([\gamma])_i = \int_\gamma \eta_i = \int_{\tilde{x}_0}^{\gamma \cdot \tilde{x}_0} d\phi_i = \phi_i(\gamma \cdot \tilde{x}_0) - \phi_i(\tilde{x}_0) \stackrel{(*)}{\in} \Lambda.$$

Hence A descends to a homomorphism of complex tori

$$\tilde{\varphi} : \text{Alb}(X) = H^0(X, \Omega_X^1)^\vee / H_1(X, \mathbb{Z}) \longrightarrow \mathbb{C}^g / \Lambda = T.$$

For $x \in X$, lift a path from x_0 to x to a path in $\tilde{X}$ from $\tilde{x}_0$ to some $\tilde{x}$ over x . Then

$$\varphi(x) - \varphi(x_0) = \phi(\tilde{x}) - \phi(\tilde{x}_0) \bmod \Lambda = \left(\int_{x_0}^x \eta_1, \dots, \int_{x_0}^x \eta_g \right) \bmod \Lambda = A(\text{alb}_X(x)) = \tilde{\varphi}(\text{alb}_X(x)),^a$$

using the fundamental theorem of calculus on each component and the definition of A . This gives

$$\varphi(x) = \tilde{\varphi}(\text{alb}_X(x)) + \varphi(x_0), \quad x \in X.$$

■

^aHere we abuse notation slightly: $A(\text{alb}_X(x))$ means applying A to one of the representatives of the Albanese class in $H^0(X, \Omega_X^1)^\vee$. Since A maps $H_1(X, \mathbb{Z})$ into the lattice Λ , this abuse is harmless.

The Albanese mapping has a relative version.

Theorem 2.5. *Let X be a compact manifold in $\mathcal{C}$, and let $f : X \rightarrow Y$ be a fiber space. Then an Albanese reduction of f exists and is unique up to bimeromorphic equivalence. Here $A(X/Y) \rightarrow Y$ is a torus fibration whose general fibers are $\text{Alb}(X_y)$.*

$$\begin{array}{ccc}
 X & \xrightarrow{\quad} & A(X/Y) \\
 & \searrow \quad \swarrow & \\
 & Y &
 \end{array}$$

The restriction to a general fiber X_y gives the Albanese reduction of X_y .

2.3 The Albanese generation lemma

In general the Albanese map need not have connected fibers, nor need it be surjective. Although it is not surjective in general, its image nonetheless generates the Albanese variety; we call this the Albanese Generation Lemma. We then give several applications.

Proposition 2.6 ([Uen75, Lemma 9.14], Albanese Generation Lemma). *The image $\text{alb}(V)$ generates the torus $\text{Alb } V$ as a group. More precisely, for sufficiently large k , the morphism*

$$\text{alb}_V^k : V^k \rightarrow \text{Alb } V, \quad (z_1, \dots, z_k) \mapsto \sum_i \text{alb}_V(z_i),$$

where the right-hand sum is taken using the group structure of the torus $\text{Alb } V$, is surjective.

Proof. Due to limited time, we omit the complete proof and only sketch the idea. After translating the base point, we may assume that there exists a point $z \in V$ with $\alpha(z) = 0$. For each $k \geq 1$ the image gives a subvariety $A_k \subset \text{Alb } V$, and these form a nested sequence

$$A_1 \subset A_2 \subset \dots \subset A_n \subset \dots$$

Since each A_k is irreducible^a, there exists an n_0 such that

$$A_{n_0} = A_{n_0+1} = \dots = A.$$

The difficult part is to show that A is a subtorus (i.e. stable under the group law); we omit this.

We end the proof by showing that $\text{Alb}(V) = A$ using the universal property. Indeed, let $g : V \rightarrow B$ be a morphism into a complex torus B ; then by the universal property (Theorem 2.4), g factors through the Albanese mapping $\alpha : V \rightarrow \text{Alb}(V)$. Since the image of the Albanese mapping is contained in A , we have the following commutative diagram.

$$\begin{array}{ccccc}
 V & \xrightarrow{\quad g \quad} & B \\
 & \searrow \quad \swarrow & \\
 & \text{Alb } V & \\
 \alpha \swarrow & \uparrow & \searrow h \\
 & A &
 \end{array}$$

Since A is a complex torus, the universal property ([Theorem 2.4](#)) again shows that A is the Albanese torus of V , that is, $A = \text{Alb}(V)$. ■

^aYL: here irreducibility is used; the general fact is that if $X \subset Y$ are irreducible subvarieties, then either $X = Y$ or $\dim X < \dim Y$; see e.g. Vakil's FOAG 12.1.5, whose proof also works in the analytic case.

As a direct consequence of the previous proposition, we see that for a Moishezon manifold the Albanese torus is an abelian variety. For a general compact Kähler manifold, however, the Albanese variety need not be projective.

Corollary 2.7 ([Uen75, Proposition 9.15]). *If X is a Moishezon manifold (in particular, if X is projective), then $\text{Alb}(X)$ is an abelian variety.*

Proof. There is an integer $n > 0$ such that the n -fold sum map

$$\text{alb}_X^n : X \times \cdots \times X \longrightarrow \text{Alb}(X), \quad (z_1, \dots, z_n) \mapsto \text{alb}_X(z_1) + \cdots + \text{alb}_X(z_n),$$

is surjective: the translates of $\text{alb}_X(X)$ generate $\text{Alb}(X)$ as a group. Since X is Moishezon, so is the product $X \times \cdots \times X$. The map alb_X^n is a proper surjective holomorphic map from this Moishezon manifold onto $\text{Alb}(X)$, and a proper surjective holomorphic image of a Moishezon space is again Moishezon; hence $\text{Alb}(X)$ is Moishezon.

Finally, a compact complex torus is Moishezon if and only if it is an abelian variety, since a compact complex torus is always Kähler and a Kähler Moishezon manifold is always projective. ■

Using the Albanese generation lemma, we can also give a criterion for the Albanese map to be surjective in terms of the Kodaira dimension of its image. We first state, without proof, a lemma on the Kodaira dimension of subvarieties of a complex torus.

Lemma 2.8 ([Uen75, Corollary 10.5], Kodaira dimension of subvarieties of complex torus).

Let $B \subset A$ be a subvariety of a complex torus. Then:

- (1) $\kappa(B) \geq 0$,
- (2) $\kappa(B) = 0$ if and only if B is a translate of a subtorus of A ;
- (3) if B generates A as a group (under translations), then $\kappa(B) > 0$.

Corollary 2.9 ([Uen75, Corollary 10.6]). *Let $\alpha : V \rightarrow \text{Alb}(V)$ be the Albanese mapping. Then $\kappa(\alpha(V)) \geq 0$, with equality to 0 if and only if α is surjective.*

Proof. Let $Z = \alpha(V) \subseteq \text{Alb}(V)$. By the Albanese generation lemma, Z generates $\text{Alb}(V)$ as a group. If α is not surjective, Z is a proper subvariety generating $\text{Alb}(V)$, so $\kappa(Z) > 0$ by Lemma 2.8. If α is surjective, $Z = \text{Alb}(V)$ has trivial canonical bundle, so $\kappa(Z) = 0$. In either case $\kappa(Z) \geq 0$, with equality precisely when α is surjective. ■

A variety with maximal Albanese dimension and canonical singularities has non-negative

Kodaira dimension.

Lemma 2.10. *Let X be a Kähler variety with maximal Albanese dimension, then the Kodaira dimension $\kappa(X) \geq 0$.*

Proof. Since the Kodaira dimension is a birational invariant for varieties with canonical singularities, we may assume that X is smooth. Then $\kappa(X) = \kappa(\alpha(X)) \geq 0$, since a subvariety of the complex torus $A(X)$ has non-negative Kodaira dimension. ■

2.4 Albanese mapping for varieties with rational singularities

For applications to the minimal model program one needs the Albanese map for mildly singular (not necessarily smooth) varieties. We first show a birational invariance property of the Albanese mapping.

Lemma 2.11 ([Uen75, Lemma 9.11]). *A meromorphic map from a compact complex manifold to a complex torus is automatically a morphism (holomorphic everywhere, with no points of indeterminacy).*

Proof. Take a common resolution of the meromorphic map $X \dashrightarrow Y$, given by $p : W \rightarrow X$ and $q : W \rightarrow Y$. Suppose the exceptional set of p is non-empty. By Abhyankar's lemma there exists a rational curve $C \subset \text{Ex}(p)$, which is therefore contracted by p . On the other hand Y contains no rational curve, so C is contracted by q as well — which is impossible for a curve in a common resolution. Hence the exceptional set of p is empty, and $X \dashrightarrow Y$ is in fact a morphism. ■

Proposition 2.12 (Birational invariance; [Uen75, Proposition 9.12]). *If $f : \widehat{X} \rightarrow X$ is a bimeromorphic morphism of complex manifolds, then $(\text{Alb}(X), \text{alb}_X \circ f)$ is the Albanese torus of $\widehat{X}$. Consequently, having maximal Albanese dimension is a birational invariant, and one may speak of the Albanese dimension of a singular or non-complete variety via any (equivalently, every) resolution.*

Proof. Write $\alpha = \text{alb}_X$, $\widehat{\alpha} = \text{alb}_{\widehat{X}}$, and let $g : X \dashrightarrow \widehat{X}$ be the inverse of f . Fix $\widehat{x}_0 \in \widehat{X}$ and translate by constant of $\alpha, \widehat{\alpha}$ so that $\widehat{\alpha}(\widehat{x}_0)$ and $\alpha(f(\widehat{x}_0))$ are the respective origins. By Lemma 2.11, the meromorphic map $\alpha \circ f : \widehat{X} \rightarrow \text{Alb}(X)$ is in fact a morphism, so the universal property (Theorem 2.4) produces a homomorphism of complex tori

$$A(f) : \text{Alb}(\widehat{X}) \rightarrow \text{Alb}(X), \quad \alpha \circ f = A(f) \circ \widehat{\alpha}.$$

Symmetrically, $\widehat{\alpha} \circ g : X \dashrightarrow \text{Alb}(\widehat{X})$ is, again by Lemma 2.11, a genuine morphism, giving a homomorphism

$$A(g) : \text{Alb}(X) \rightarrow \text{Alb}(\widehat{X}), \quad \widehat{\alpha} \circ g = A(g) \circ \alpha.$$

Since $f \circ g = \text{id}_X$ and $g \circ f = \text{id}_{\widehat{X}}$ as bimeromorphic maps, substituting the two displayed identities into each other gives $A(f) \circ A(g) = \text{id}$ and $A(g) \circ A(f) = \text{id}$ on the dense open subsets where the relevant compositions are defined, and hence everywhere. So $A(f)$ is an isomorphism, and $(\text{Alb}(X), \alpha \circ f)$ satisfies the universal property of the Albanese torus for $\widehat{X}$. ■

Definition 2.13 (Albanese map for singular varieties). Let X be a compact normal variety in Fujiki class $\mathcal{C}$. Let $\mu : Y \rightarrow X$ be a Kähler resolution of X . We then define the Albanese mapping as

$$\text{alb}_Y \circ \mu^{-1} : X \dashrightarrow \text{Alb}(Y).$$

This is independent of the choice of resolution by [Proposition 2.12](#), hence well defined.

Typically, we will work with Fujiki varieties with rational singularities, as in this case the Albanese mapping is holomorphic.

Proposition 2.14 ([BS95, Lemma 2.4.1]). Let X be a complete normal variety in Fujiki's class $\mathcal{C}$ with at worst rational singularities. Then the Albanese map $X \dashrightarrow \text{Alb}(X)$ defined via a resolution is in fact an everywhere-defined morphism $\text{alb}_X : X \rightarrow \text{Alb}(X)$, and coincides with the natural map $X \rightarrow \text{Pic}^0(X)^\vee$ induced by the Poincaré line bundle.

Before proving this proposition, let us say a few words about the natural holomorphic map $X \rightarrow \text{Pic}^0(X)^\vee$. Recall that, given a compact normal variety in Fujiki class $\mathcal{C}$, there is a universal line bundle $\mathcal{P}$ on $X \times \text{Pic}^0(X)$ such that

$$\mathcal{P}|_{X \times [L]} \cong L, \quad \mathcal{P}|_{\{x_0\} \times \text{Pic}^0(X)} \cong \mathcal{O}_{\text{Pic}^0(X)}.$$

We can then consider the restriction $\mathcal{P}|_{\{x\} \times \text{Pic}^0(X)} \in \text{Pic}(\text{Pic}^0(X)) = \text{Pic}^0(X)^\vee$, which clearly varies holomorphically in x and hence defines a holomorphic map

$$X \rightarrow \text{Pic}^0(X)^\vee, \quad x \mapsto \mathcal{P}_x.$$

Proof. Let $\mu : Y \rightarrow X$ be a resolution and set $\text{Alb}(X) := \text{Alb}(Y)$. Rational singularities give $H^1(X, \mathcal{O}_X) \cong H^1(Y, \mathcal{O}_Y)$ via μ^* (a Leray spectral sequence argument), from which one deduces that $\mu^* : \text{Pic}^0(X) \rightarrow \text{Pic}^0(Y)$ is étale^a and injective (as μ is a birational morphism of normal varieties), hence μ^* is an isomorphism. Since Y is smooth and Kähler, we have

$$\text{Alb}(Y) \cong \text{Pic}^0(Y)^\vee \cong \text{Pic}^0(X)^\vee.$$

Hence the Albanese map is nothing but the natural holomorphic map

$$X \rightarrow \text{Pic}^0(X)^\vee$$

mentioned above. ■

^aTo see this, take the differential to get

$$d\mu^* : H^1(X, \mathcal{O}_X) \rightarrow H^1(Y, \mathcal{O}_Y),$$

which is an isomorphism, using the usual identification of the tangent space of the Picard variety with $H^1(X, \mathcal{O}_X)$.

3 Irregularity and Albanese dimension

We now introduce the numerical invariants attached to $\text{Alb}(X)$ and alb_X , following Ueno's book [Uen75], which is the standard reference for this material.

Definition 3.1 (Irregularity; [Uen75, Definition 9.20]). Let V be a complex variety with non-singular model V^* . The irregularity of V is

$$q(V) := \dim_{\mathbb{C}} H^1(V^*, \mathcal{O}_{V^*}),$$

independent of the choice of resolution V^* and a bimeromorphic invariant of V .

Definition 3.2 (Albanese dimension; [Uen75, Definition 9.21]). The Albanese dimension of a complex variety V is $a(V) := \dim \text{Alb}(V^*)$, the dimension of the Albanese torus of a non-singular model V^* . We say V has maximal Albanese dimension if $a(V) = \dim V$, equivalently the Albanese map is generically finite onto its image.

Definition 3.3 (Augmented irregularity).

$$\widehat{q}(X) := \sup \{ q(X') \mid X' \rightarrow X \text{ a finite étale cover} \} \in \mathbb{Z}_{\geq 0} \cup \{\infty\}.$$

Proposition 3.4 (Behaviour under surjective morphisms, [Uen75, Lemma 9.23]). Let $f : V \rightarrow W$ be a surjective morphism of complex varieties. Then

$$q(V) \geq q(W), \quad a(V) \geq a(W).$$

Proof. Pullback of forms $f^* : H^0(W, \Omega_W^1) \rightarrow H^0(V, \Omega_V^1)$ is injective because f is dominant, which gives $q(V) \geq q(W)$. The Albanese-dimension inequality follows by applying this together with the universal property (Theorem 2.4): the composite $V \rightarrow W \rightarrow \text{Alb}(W)$ factors through $\text{Alb}(V)$, so $\text{Alb}(W)$ is (a quotient of) a further quotient torus of $\text{Alb}(V)$. ■

We next prove a subadditivity inequality using the E_2 -degeneration of a spectral sequence. This relation will be important in the next section.

Lemma 3.5 (Subadditivity of irregularity). Let $g : X \rightarrow Y$ be a smooth fibration with connected fibres between compact Kähler manifolds, with fibre F . Then

$$q(X) \leq q(Y) + q(F),$$

with equality if and only if the monodromy representation $\pi_1(Y) \rightarrow \text{Aut}(H^1(F, \mathbb{Z}))$ is trivial.

As with the Iitaka conjecture, the subadditive direction requires no extra hypothesis, whereas for the additive relation a Hodge-theoretic condition appears. Later we will upgrade this condition to augmented irregularity.

Proof. Consider the Leray spectral sequence of the constant sheaf $\mathbb{R}_X$ for g ,

$$E_2^{s,t} = H^s(Y, R^t g_* \mathbb{R}_X) \implies H^{s+t}(X, \mathbb{R}).$$

Since the fibres are connected and g is (differentiably) a locally trivial fibre bundle, $R^0 g_* \mathbb{R}_X = \mathbb{R}_Y$, and $R^1 g_* \mathbb{R}_X$ is a locally constant real vector bundle of rank $2q(F)$ containing $R^1 g_* \mathbb{Z}_X$ as a lattice, so its monodromy lies in $SL_{2q(F)}(\mathbb{Z})$. Indeed $H^1(F, \mathbb{Z})$ is a free lattice of rank $b_1 = 2q(F)$, so the monodromy lies in $GL(H^1(F, \mathbb{Z})) = GL_{2q(F)}(\mathbb{Z})$. On the other hand, for a geometric family the monodromy preserves the intersection pairing, hence actually lies in the smaller subgroup $SL_{2q(F)}(\mathbb{Z})$.

Since the global sections of a local system are isomorphic to the monodromy-invariant part of the fiber, we get

$$\dim H^0(Y, R^1 g_* \mathbb{R}_X) \leq 2q(F),$$

with equality iff the monodromy is trivial. By Blanchard's theorem, the Leray spectral sequence of a Kähler fibration degenerates at E_2^a , so

$$\begin{aligned} 2q(X) &= \dim H^1(X, \mathbb{R}) \\ &= \dim H^1(Y, \mathbb{R}) + \dim H^0(Y, R^1 g_* \mathbb{R}_X) \\ &\leq 2q(Y) + 2q(F), \end{aligned}$$

with equality iff the monodromy representation $\pi_1(Y) \rightarrow \text{Aut}(H^1(F, \mathbb{Z}))$ is trivial. ■

^aOne should be careful: the Kähler condition is vital for the spectral-sequence argument; a non-Kähler smooth fibration may fail to have E_2 -degeneration, e.g. the Hopf fibration.

Lemma 3.6 (Subadditive for augmented irregularity). *In the situation of Lemma 3.5, $\widehat{q}(X) \leq \widehat{q}(Y) + \widehat{q}(F)$.*

Proof. The homotopy exact sequence of the fibration gives

$$\cdots \rightarrow \pi_2(Y) \rightarrow \pi_1(F) \rightarrow \pi_1(X) \rightarrow \pi_1(Y) \rightarrow 1.$$

A finite étale cover $\widetilde{X}$ of X corresponds to a finite-index subgroup $\pi_1(\widetilde{X}) \subset \pi_1(X)$; its image in $\pi_1(Y)$ and preimage in $\pi_1(F)$ are again finite-index, giving finite étale covers $\widetilde{Y}, \widetilde{F}$ fitting into a smooth fibration $\widetilde{X} \rightarrow \widetilde{Y}$ with fibre $\widetilde{F}$.

$$\begin{array}{ccc}
\tilde{F} & \longrightarrow & F \\
\downarrow & & \downarrow \\
\tilde{X} & \longrightarrow & X \\
\downarrow & & \downarrow \\
\tilde{Y} & \longrightarrow & Y
\end{array}$$

By Lemma 3.5,

$$q(\tilde{X}) \leq q(\tilde{Y}) + q(\tilde{F}) \leq \widehat{q}(Y) + \widehat{q}(F);$$

taking the supremum over $\tilde{X}$ gives the claim. $\blacksquare$

We next prove a rigidity theorem for torus fibrations using Hodge theory.

Lemma 3.7 (Rigidity of Kähler torus fibrations). *A smooth Kähler fibration by complex tori*

$$f : X \rightarrow \mathbb{C}$$

is holomorphically trivial.

Proof. By a result of Griffiths and Schmid, the classifying (period) map $\mathbb{C} \rightarrow D$ into the Siegel upper half space — the moduli space of complex tori polarized by a Kähler form ω , with given periods — is always constant, by the Brody hyperbolicity of D . $\blacksquare$

Similar to the Iitaka additivity conjecture, the subadditive direction does not require Hodge theory. The (super)additive direction, however, requires some Hodge theory.

Proposition 3.8 (Demailly–Peternell–Schneider’s addition formula for irregularity, [DPS94, Proposition 3.12]). *Let $g : X \rightarrow Y$ be a smooth fibration with connected fibres between compact Kähler manifolds, with fibre F . Then*

$$q(X) \leq q(Y) + q(F), \quad \widehat{q}(X) \leq \widehat{q}(Y) + \widehat{q}(F).$$

If moreover Y is a complex torus and $\pi_1(F)$ is virtually abelian with the Albanese map of every finite étale cover of F of constant rank, then $\widehat{q}(X) = \widehat{q}(Y) + \widehat{q}(F)$.

Proof. We only highlight the idea of the proof. Consider the relative Albanese factorization $X \rightarrow A(X/Y) \rightarrow Y$, a smooth torus fibration with fibres $A(F_y)$. We show that $A(X/Y) \rightarrow Y$ is a holomorphically locally trivial torus bundle, using the rigidity result proved above: since Y is a complex torus, its universal cover is $\mathbb{C}^n$, which is why we can apply the Griffiths–Schmid rigidity result^a.

for a small $U \ni y$, on $g^{-1}(U)$ there are holomorphic maps over U

$$g^{-1}(U) \rightarrow A(g^{-1}(U)/U) \simeq A(F_y) \times U \xrightarrow{\text{pr}_2} A(F_y).$$

Recall that we already showed that

$$\rho : \pi_1(Y) \rightarrow GL(H^1(F_y, \mathbb{R})) = GL_{2q}(\mathbb{R})$$

has image $\rho(\pi_1(Y)) \subseteq SL_{2q}(\mathbb{Z})$, a discrete subgroup. We now introduce a compactness constraint: we construct a metric such that

$$\langle \rho(\gamma)u, \rho(\gamma)v \rangle_y = \langle u, v \rangle_y.$$

To do this we take holomorphic sections on F_y so that

$$\langle u, u \rangle_y = \int_{F_y} iu' \wedge \bar{u}' \wedge \omega^{p-1} > 0.$$

Since the torus fibration is holomorphically trivial, we may take u', v' to be d -closed holomorphic 1-forms on all of $g^{-1}(U)$; restricting to each fibre $F_{y'}$, define

$$\langle u, u \rangle_{y'} = \int_{F_{y'}} \Theta, \quad \Theta := i u' \wedge \bar{v}' \wedge \omega^{p-1}, \quad p = \dim F_{y'}.$$

By Stokes' formula we can check

$$\int_{F_{y_1}} \Theta - \int_{F_{y_0}} \Theta = \int_{g^{-1}(\gamma)} d\Theta = 0,$$

hence it is independent of y .

In summary, for the Kähler fibration $X \rightarrow Y$ with fibre F , the monodromy

$$\pi_1(Y) \rightarrow \text{Aut}(H^1(F', \mathbb{Z}))$$

has finite image, since it is contained in $SL_{2q}(\mathbb{Z}) \cap O_{2q}(\langle, \rangle)$. Hence, after a finite cover giving a fibration $\tilde{X} \rightarrow \tilde{Y}$ with fibre $\tilde{F} = F'$ and trivial monodromy, we apply [Lemma 3.5](#) to deduce the equality. ■

^aTo apply the Griffiths–Schmid result we must check that $A(X/Y)$ is Kähler; we omit this and refer to the original proof.

4 Albanese fibrations for Kähler manifolds with nef tangent bundle

We now turn to Demailly–Peternell–Schneider's classification of compact Kähler manifolds with numerically effective (nef) tangent bundle, in which the Albanese fibration is the central structural device. Their remarkable work shows that for a compact Kähler manifold with T_X nef, alb_X is a smooth fibration whose fibres are Fano (hence simply connected).

First we show a result particularly suited to the induction.

Theorem 4.1 (Demailly–Peternell–Schneider first structure theorem, Proposition 3.9). *Let X be a Kähler manifold with nef tangent bundle T_X , and let $\alpha: X \rightarrow A(X)$ be the Albanese map, where $A(X) = \mathbb{C}^q / \Lambda$ is the Albanese torus. Then:*

1. α is a submersion;
2. the relative tangent bundle $T_{X/A(X)}$ is nef;
3. the fibres of α are connected and have nef tangent bundle.

Proof. Step 1: α is a submersion. The torus $A(X) = \mathbb{C}^q / \Lambda$ carries translation-invariant holomorphic 1-forms $dz_1, \dots, dz_q$. Pulling back via α gives global sections

$$u_i = \alpha^*(dz_i) = d\alpha_i \in H^0(X, \Omega_X^1), \quad i = 1, \dots, q,$$

so that the differential of α is encoded by

$$d\alpha = (d\alpha_1, \dots, d\alpha_q): T_X \longrightarrow \alpha^*T_{A(X)}.$$

Suppose α fails to be a submersion at some point $x \in X$, i.e. $\text{rk}(d\alpha(x)) < q$. Then the vectors $(d\alpha_i(x))_{i=1, \dots, q}$ are linearly dependent, so there is a nonzero linear combination

$$u = \lambda_1 u_1 + \dots + \lambda_q u_q \in H^0(X, \Omega_X^1)$$

with $u(x) = 0$.

Since T_X is nef, its dual bundle $\Omega_X^1 = T_X^*$ is a dual of a nef, weakly positively curved vector bundle. Since a nontrivial section of the dual of a nef vector bundle vanishes nowhere, u cannot vanish at any point of X . This contradicts $u(x) = 0$. Hence $\text{rk}(d\alpha) = q$ everywhere, and α is a submersion.

Step 2: the fibres have nef tangent bundle. Since α is a submersion, we have the short exact sequence of vector bundles on X

$$0 \longrightarrow T_{X/A(X)} \longrightarrow T_X \xrightarrow{d\alpha} \alpha^*T_{A(X)} \longrightarrow 0.$$

As $A(X)$ is a torus, $T_{A(X)}$ is trivial, hence $\alpha^*T_{A(X)}$ is nef (indeed trivial). Since T_X is nef and quotients of nef bundles are nef, while the kernel of a surjection from a nef bundle onto a nef bundle is again nef, it follows that $T_{X/A(X)}$ is nef. Restricting to a fibre $F = \alpha^{-1}(y)$ gives $T_{X/A(X)}|_F = T_F$, so every fibre has nef tangent bundle.

Step 3: the fibres are connected. Suppose some fibre of α is disconnected. Taking connected components of fibres yields the Stein factorization

$$\alpha: X \xrightarrow{f} Y \xrightarrow{g} A(X),$$

where f has connected fibres and, since α is a submersion, f is a smooth fibration; hence g is a finite étale covering^a. We may assume $\deg(g) > 1$, since otherwise g is an isomorphism and the fibres of α are already connected.

Because $g: Y \rightarrow A(X)$ is a finite étale cover of a complex torus, Y is itself a complex torus.

By the universal property of the Albanese map applied to $f: X \rightarrow Y$, there is a factorization

$$X \xrightarrow{\alpha} A(X) \xrightarrow{h} Y$$

with $g \circ h = \text{id}_{A(X)}$ and $h \circ g = \text{id}_Y$ up to translation, by the uniqueness part of the universal property of complex tori applied to the composite $A(X) \rightarrow Y \rightarrow A(X)$. In particular g is an isomorphism, since g is a finite morphism between abelian varieties of the same dimension whose tangent map is an isomorphism, hence an isogeny of degree 1. This contradicts $\deg(g) > 1$. Therefore the fibres of α are connected, completing the proof. ■

^aThis is a smooth version of Stein factorization, that is if α is smooth, we can assume that g is étale, see this version e.g. in Shin-ichi's paper.

Using the subadditivity of irregularity, we can establish the finer structure of the Albanese fibration, namely that the fibres are Fano manifolds. First we briefly mention two facts about Fano manifolds.

Theorem 4.2. *A Fano manifold is simply connected. (More generally, rationally connected varieties with klt singularities are simply connected).*

It is worth mentioning that singularities may break simple-connectedness, e.g. a nodal rational curve is not simply connected.

In the proof we use the following Fano criterion in terms of irregularity.

Theorem 4.3. *Let X be a Kähler manifold with T_X nef. One of the two following situations occurs:*

- (i) $c_1(X)^n > 0$. Then X is a Fano manifold (that is, a projective algebraic manifold with K_X^{-1} ample) and $\chi(X, \mathcal{O}_X) = 1$.
- (ii) $c_1(X)^n = 0$. Then $\chi(X, \mathcal{O}_X) = 0$ and X admits a non trivial holomorphic p -form of odd degree. Moreover, there is a finite étale cover $\tilde{X}$ of X such that

$$q(\tilde{X}) := h^0\left(X, \Omega_{\tilde{X}}^1\right) > 0.$$

The proof is hard and we omit it. With these two facts about Fano geometry at hand, we can now finish the proof of the DPS main result.

Theorem 4.4 (Demailly–Peters–Schneider second structure theorem, [DPS94, Theorem 3.14, 3.15]). *Let X be a compact Kähler manifold with T_X nef, and let $\tilde{X} \rightarrow X$ be a finite étale cover realising the maximal irregularity $q = \tilde{q}(X) = q(\tilde{X})$. Then:*

- (i) $\pi_1(\tilde{X}) \cong \mathbb{Z}^{2q}$;
- (ii) *the Albanese map $\text{alb}_{\tilde{X}} : \tilde{X} \rightarrow \text{Alb}(\tilde{X})$ is a smooth fibration over a q -dimensional torus, with nef relative tangent bundle;*
- (iii) *most importantly the fibres F of this fibration are Fano manifolds with nef tangent bundle.*

Consequently $\pi_1(X)$ is, as an abstract group, an extension of a finite group by $\mathbb{Z}^{2q}$.

Proof. The two statements (main theorem for $\tilde{X}$, and the fundamental group extension for X) are proved simultaneously by induction on $n = \dim X$. Assume both hold in dimension $< n$.

Step 1 (Albanese fibration is a smooth fibration over a torus). Let $\tilde{X} \rightarrow X$ be a finite étale cover realizing the maximal irregularity $q = q(\tilde{X})$. Applying 4.1 (the Albanese map of a Kähler manifold with nef tangent bundle is a submersion onto its Albanese torus, with nef relative tangent bundle and connected fibres) to $\tilde{X}$ gives (ii) directly: $\alpha : \tilde{X} \rightarrow A(\tilde{X})$ is a smooth fibration over a q -dimensional torus with nef relative tangent bundle, and the fibres F inherit nef tangent bundle.

By the induction hypothesis applied to F (which has dimension $< n$), the fundamental group $\pi_1(F)$ contains an abelian subgroup of finite index.

Step 2 (Fibers are Fano manifolds). The irregularity is subadditive along smooth Kähler fibrations:

$$\tilde{q}(\tilde{X}) = \tilde{q}(A(\tilde{X})) + \tilde{q}(F).$$

Since $\tilde{X}$ already realizes the maximal irregularity, one has $\tilde{q}(\tilde{X}) = q(\tilde{X}) = \dim A(\tilde{X})$ (the irregularity is achieved, so it equals its own "generic" value and cannot grow further under étale covers). Comparing the two displays forces

$$\tilde{q}(F) = 0.$$

Feeding this into the Fano criterion for Kähler manifolds with nef tangent bundle and non-vanishing first Chern class / Euler characteristic (4.3) rules out the "non-Fano" alternative, so F is a Fano manifold with nef tangent bundle. This proves (iii).

Step 3 (fundamental group of $\tilde{X}$). Fano manifolds are simply connected^a, so $\pi_1(F) = 1$. The homotopy long exact sequence of the fibration $F \rightarrow \tilde{X} \rightarrow A(\tilde{X})$ collapses to

$$1 = \pi_1(F) \longrightarrow \pi_1(\tilde{X}) \longrightarrow \pi_1(A(\tilde{X})) \longrightarrow 1,$$

giving $\pi_1(\tilde{X}) \cong \pi_1(A(\tilde{X})) \cong \mathbb{Z}^{2q}$, using that the fundamental group of a complex torus is its lattice. This proves (i).

Step 4 (descending to X , the Galois covering trick). The issue is that for the finite étale cover $\tilde{X} \rightarrow X$, the induced fundamental group need not be a normal subgroup, as required in the theorem. Hence we pass to a further covering $\tilde{\tilde{X}} \rightarrow \tilde{X} \rightarrow X$ that is Galois over X , which restores the normality condition. ■

^aSimply connectedness of Fano manifold is highly non-trivial result, see my appendix.

5 Varieties with maximal Albanese dimension

Next we discuss a class of varieties with very nice birational geometry: those of maximal Albanese dimension. Our presentation follows [Fuj13] and [?].

Varieties of maximal Albanese dimension have very good minimal model theory, because the target of the Albanese map is (a subvariety of) an abelian variety, and abelian varieties contain no rational curves. The following lemma is the precise mechanism.

Lemma 5.1 ([Fuj13, Lemma 3.2, 3.3]). *Let (X, Δ) be a log canonical pair and $f : X \rightarrow S$ a surjective morphism of projective varieties, with S containing no rational curves. If $K_X + \Delta$ is f -nef, then it is nef. Consequently, if (X', Δ') is a minimal model of (X, Δ) over S , it is already a minimal model of (X, Δ) .*

Proof. If $K_X + \Delta$ were not nef, the cone theorem would produce a rational curve $C \subset X$ with $(K_X + \Delta) \cdot C < 0$. Since $K_X + \Delta$ is f -nef, C cannot be f -vertical, so $f(C)$ is a positive-dimensional subvariety of S . But $f(C)$ is the image of $\mathbb{P}^1$, hence a rational curve, contradicting that S has none. The second statement is the same argument applied to $K_{X'} + \Delta'$, which is f -nef by definition of a relative minimal model. ■

Remark 5.2. Abelian varieties, and more generally subvarieties of abelian varieties, contain no rational curves (a non-constant map $\mathbb{P}^1 \rightarrow A$ would lift to a non-constant holomorphic map $\mathbb{P}^1 \rightarrow \mathbb{C}^g$, impossible by compactness of $\mathbb{P}^1$). So Lemma 5.1 applies with S any subvariety of $\text{Alb}(X)$, in particular with $S = a_X(X)$.

Theorem 5.3 (Fujino; [Fuj13, Theorem 4.3]). *Let (X, Δ) be a projective klt pair of maximal Albanese dimension. Then (X, Δ) has a good minimal model.*

Proof. Klt singularities are in particular rational, so Proposition 2.14 shows $a_X : X \rightarrow \text{Alb}(X)$ is an everywhere-defined morphism; let $S := a_X(X)$ and $f := a_X : X \rightarrow S$. Since X has maximal Albanese dimension, f is generically finite, hence $K_X + \Delta$ is f -big (a generically finite morphism makes every $\mathbb{R}$ -divisor relatively big). By the relative existence theorem for minimal models of big divisors [BCHM10, Theorem 1.2], there is a minimal model $(X', \Delta') \rightarrow S$ of (X, Δ) over S .

Since $S \subseteq \text{Alb}(X)$, Lemma 5.1 shows (X', Δ') is already an (absolute) minimal model

of (X, Δ) : $K_{X'} + \Delta'$ is nef. Finally, by the abundance theorem for varieties of maximal Albanese dimension, (X', Δ') is a good minimal model. ■

A general subvariety of a variety with maximal Albanese dimension again has maximal Albanese dimension.

Lemma 5.4. *Let X be a projective variety with maximal Albanese dimension. Let $\pi : \bar{X} \rightarrow X$ be a resolution and let $\alpha : \bar{X} \rightarrow \text{Alb}(\bar{X})$ be the Albanese mapping. Let Y be a subvariety of X . Assume that $Y \not\subset \pi(\text{Exc}(\pi) \cup \text{Exc}(\beta))$, where $\beta : \bar{X} \rightarrow V$ is the Stein factorization of $\alpha : \bar{X} \rightarrow \alpha(\bar{X})$. Then Y has maximal Albanese dimension.*

We next prove the abundance theorem for varieties with maximal Albanese dimension.

Theorem 5.5 (Abundance conjecture for varieties with maximal Albanese dimension, [Fuj13]). *Let (X, Δ) be a projective klt pair such that X has maximal Albanese dimension. Assume that $K_X + \Delta$ is nef. Then $K_X + \Delta$ is semi-ample.*

Proof. We prove the result by induction on dimension. By taking a small $\mathbb{Q}$ -factorial modification and a terminalization, we may assume that (X, Δ) is a $\mathbb{Q}$ -factorial terminal pair. By the previous discussion, since X has maximal Albanese dimension, $\kappa(K_X + \Delta) \geq 0$. The key point is that it suffices to show $\kappa(K_X + \Delta) > 0$. Indeed, once we know this, we argue as in [?]: there exist nonsingular varieties Y, Z , a birational morphism $\mu : Y \rightarrow X$, and the Iitaka fibration $f : Y \rightarrow Z$ of X , as in the following commutative diagram:

$$\begin{array}{ccc} & Y & \\ \mu \swarrow & & \searrow f \\ X & & Z \end{array}$$

Since $\kappa(K_X + \Delta) > 0$, we have $\dim Z > 0$, so a general fiber F of f has dimension $\dim F < \dim X$; on the other hand it still has maximal Albanese dimension (by Lemma 5.4). We then run the MMP on F to get

$$F \dashrightarrow F',$$

where F' again has maximal Albanese dimension (cf. Proposition 2.12). Since $\dim F' < \dim X$, by induction F' satisfies log abundance; combined with $\kappa(K_F + \Delta_F) = 0$, this implies

$$K_{F'} + \Delta_{F'} \sim_{\mathbb{Q}} 0.$$

We then deduce the result directly, as in Kawamata [?] (see the appendix of the previous note for details).

Hence it remains to prove that $\nu(K_X + \Delta) > 0$ implies $\kappa(K_X + \Delta) > 0$ (weak abundance). If $\kappa(X, K_X) > 0$ then obviously $\kappa(K_X + \Delta) > 0$, so we may assume $\kappa(K_X) = 0$ and argue by contradiction. We apply [Kaw81]: in this case $X \rightarrow a(X)$ is generically finite with connected fibres, hence birational. Because K_X is effective with support equal to the

a -exceptional part,

$$\kappa(X, K_X + \Delta) \geq \kappa(X, \alpha^* \alpha_* \Delta) = \kappa(A, \alpha_* \Delta).$$

We claim $\alpha_* \Delta \neq 0$. Indeed, if $\alpha_* \Delta = 0$, then Δ is a -exceptional; since K_X is also a -exceptional, $K_X + \Delta$ is effective and a -exceptional. On the other hand, by hypothesis $K_X + \Delta$ is nef, so the negativity lemma forces $K_X + \Delta = 0$, contradicting $\nu(K_X + \Delta) > 0$. Since $\alpha_* \Delta \neq 0$, then

$$\kappa(X, K_X + \Delta) \geq \kappa(X, \alpha^* \alpha_* \Delta) = \kappa(A, \alpha_* \Delta) \geq 1,$$

where the last inequality holds because any nonzero effective divisor on an abelian variety has positive Iitaka dimension. ■

Remark 5.6 (The Kähler case, due to Das–Hacon). Fujino’s argument leans on two purely projective/algebraic tools: the relative existence theorem for minimal models of big divisors [BCHM10], and the abundance theorem for maximal Albanese dimension varieties, both available only in the projective category. Das–Hacon [DH24] prove the same conclusion for compact Kähler klt pairs (X, B) of maximal Albanese dimension, but since neither tool is available as such in the Kähler category, the proof cannot simply run the argument above analytically; instead it replaces the abundance-theorem input with a direct argument via cohomology support loci and generic vanishing (Green–Lazarsfeld–Simpson GV-sheaves) to show the relevant divisors are a_X -exceptional, combined with an induction on the Iitaka fibration to handle $\kappa > 0$. This is substantially longer and uses heavier machinery than Fujino’s projective proof, precisely because it has to reconstruct, by transcendental means, what BCHM and the abundance theorem supply for free in the projective case. Das–Hacon also record the corollary that if a_X happens to already be projective with connected fibres, one may run the $(K_X + B)$ -minimal model program directly over $\text{Alb}(X)$ (as in the proof of Theorem 5.3 above) to reach a good minimal model, without the transcendental argument.

6 Further discussion

We end this note by pointing out some further directions, in each case stated without proof.

Remark 6.1 (Conditions for the Albanese map to be a fibration). As mentioned above, in general the Albanese mapping need not be a fibration. There are, however, cases in which it is: (1) if the algebraic dimension $a(X) = 0$, see [Uen75, Lemma 13.6]; (2) If the image of alb_X is a curve C , then C is smooth of genus $\dim \text{Alb}(X)$ with connected fibres [Uen75, Proposition 9.19]; (3) When $\kappa(X) = 0$, then the Albanese map has connected fiber, see [Kaw81].

Remark 6.2 (Albanese fibrations in abundance conjectures). Using Kawamata’s structure theorem for finite covers of abelian varieties [Kaw81] to reduce the Albanese map to a fibration over an abelian variety, Nakayama proved that $\kappa_\sigma(K_X + \Delta) = 0$ implies $\kappa(K_X + \Delta) = 0$ for klt pairs [Nak04, Corollary IV.4.9].

Remark 6.3 (Albanese reduction for the Iitaka conjecture when the base has maximal Albanese dimension). Hacon–Popa–Schnell proved the Iitaka conjecture $C_{n,m}$ outright whenever the base of a fibre space has maximal Albanese dimension [HPS18].

Remark 6.4 (Good minimal models for Albanese-fibred varieties). Complementing Theorem 5.3, Lai [Lai11], showed good minimal models exist for varieties fibred by pieces that themselves admit good minimal models; specialised to the Albanese map, if $a_X : X \rightarrow \text{Alb}(X)$ is projective with connected fibres and the general fibre has a good minimal model, then so does X .

Remark 6.5 (Structure of varieties with nef anti-canonical divisor). For (X, Δ) projective klt pair, with $-(K_X + \Delta)$ nef, one can prove that alb_X is a locally constant fibration, see [Cao19, Wan22, CCM21, NW25, iMWZ25, iMW23]; the Kähler case is still open, the main difficulty being the failure of projectivity of the Albanese fibration.

Remark 6.6 (Ueno’s Conjecture K). Ueno conjectured that when $\kappa(X) = 0$, the Albanese map is not merely a fibre space (as above) but becomes, after a finite étale base change $B \rightarrow \text{Alb}(X)$, birationally trivial over B : $X \times_{\text{Alb}(X)} B$ is birational to $F \times B$ over B , where F is the general fibre, with $\kappa(F) = 0$. Ueno’s conjecture is a test case for the Iitaka conjecture.

References

- [BCHM10] C. Birkar, P. Cascini, C.D. Hacon, and J. McKernan, *Existence of minimal models for varieties of log general type*, Journal of the American Mathematical Society **23** (2010), no. 2, 405–468.
- [BS95] Mauro C. Beltrametti and Andrew J. Sommese, *The adjunction theory of complex projective varieties*, De Gruyter Expositions in Mathematics, vol. 16, Walter de Gruyter & Co., Berlin, 1995. MR 1318687
- [Cao19] Junyan Cao, *Albanese maps of projective manifolds with nef anticanonical bundles*, Ann. Sci. Éc. Norm. Supér. (4) **52** (2019), no. 5, 1137–1154. MR 4057779

- [CCM21] Frédéric Campana, Junyan Cao, and Shin-ichi Matsumura, *Projective klt pairs with nef anti-canonical divisor*, *Algebr. Geom.* **8** (2021), no. 4, 430–464. MR 4371536
- [DH24] Omprokash Das and Christopher Hacon, *Existence of good minimal models for Kähler varieties of maximal Albanese dimension*, *C. R. Math. Acad. Sci. Paris* **362** (2024), 83–91. MR 4762189
- [DPS94] Jean-Pierre Demailly, Thomas Peternell, and Michael Schneider, *Compact complex manifolds with numerically effective tangent bundles*, *J. Algebraic Geom.* **3** (1994), no. 2, 295–345. MR 1257325
- [Fuj13] Osamu Fujino, *On maximal Albanese dimensional varieties*, *Proc. Japan Acad. Ser. A Math. Sci.* **89** (2013), no. 8, 92–95. MR 3127923
- [HPS18] Christopher Hacon, Mihnea Popa, and Christian Schnell, *Algebraic fiber spaces over abelian varieties: around a recent theorem by Cao and Păun*, *Local and global methods in algebraic geometry*, *Contemp. Math.*, vol. 712, Amer. Math. Soc., [Providence], RI, [2018] ©2018, pp. 143–195. MR 3832403
- [Huy05] Daniel Huybrechts, *Complex geometry*, Universitext, Springer-Verlag, Berlin, 2005, An introduction. MR 2093043
- [iMW23] Shin ichi Matsumura and Juanyong Wang, *Structure theorem for projective klt pairs with nef anti-canonical divisor*, 2023.
- [iMWWZ25] Shin ichi Matsumura, Juanyong Wang, Xiaojun Wu, and Qimin Zhang, *Compact kähler manifolds with nef anti-canonical bundle*, 2025.
- [Kaw81] Yujiro Kawamata, *Characterization of abelian varieties*, *Compositio mathematica* **43** (1981), no. 2, 253–276.
- [Lai11] Ching-Jui Lai, *Varieties fibered by good minimal models*, *Math. Ann.* **350** (2011), no. 3, 533–547. MR 2805635
- [Nak04] Noboru Nakayama, *Zariski-decomposition and abundance*, *MSJ Memoirs*, vol. 14, Mathematical Society of Japan, Tokyo, 2004. MR 2104208
- [NW25] Philipp Naumann and Xiaojun Wu, *Albanese map for Kähler manifolds with nef anticanonical bundle*, *Math. Z.* **311** (2025), no. 2, Paper No. 25, 28. MR 4937839
- [Uen75] Kenji Ueno, *Classification theory of algebraic varieties and compact complex spaces*, *Lecture Notes in Mathematics*, vol. Vol. 439, Springer-Verlag, Berlin-New York, 1975, Notes written in collaboration with P. Cherenack. MR 506253

- [Wan22] Juanyong Wang, *Structure of projective varieties with nef anticanonical divisor: the case of log terminal singularities*, Math. Ann. **384** (2022), no. 1-2, 47–100. MR 4476228